

National University of Management

Faculty of Digital Economy

Course Title: Blockchain Technology

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Level: 2nd year

Homework Chapter 11

1. Why Consensus Exists (Centralized vs Decentralized)

In a **centralized system**, a single trusted authority controls data validation and decision-making. Because this authority is trusted, consensus is not required. Examples include banks or centralized databases where one server confirms all transactions.

In a **decentralized system**, there is no central authority. Multiple independent nodes participate in maintaining and validating the system. Since nodes do not fully trust each other, a **consensus mechanism is required** to ensure that all nodes agree on the same data, transaction order, and system state. Consensus replaces trust in a central authority with trust in a protocol.

2. Safety, Liveness, and Finality

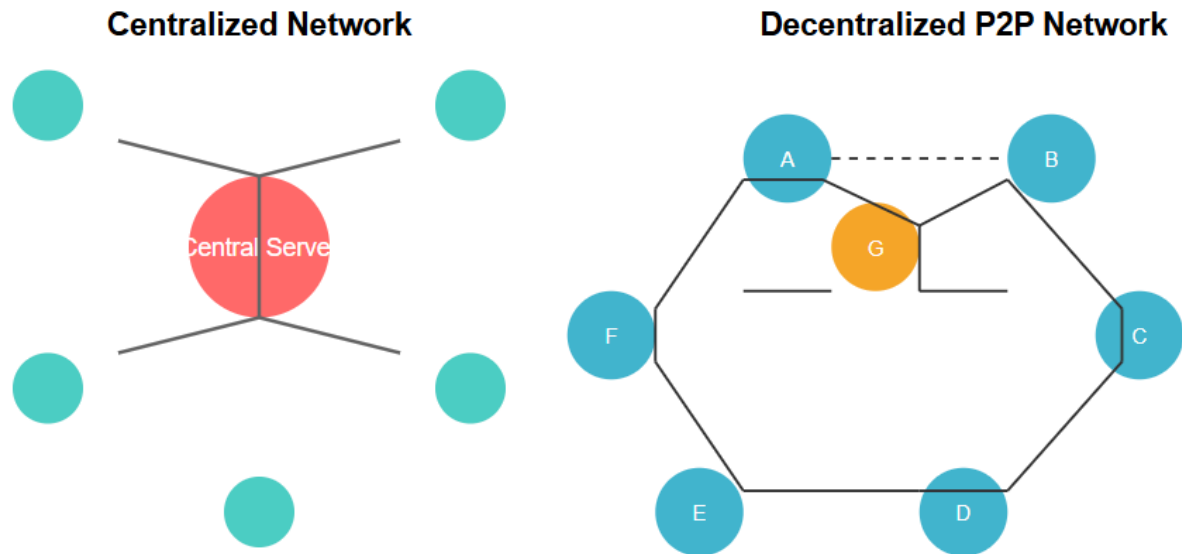
Safety refers to the property that the system never reaches an incorrect or inconsistent state. Honest nodes will not accept conflicting transactions, such as double-spending.

Liveness ensures that the system continues to make progress. Valid transactions will eventually be confirmed, and the network will not stall indefinitely.

Finality means that once a transaction is confirmed, it cannot be reversed. Some systems provide probabilistic finality, where transactions become more secure over time, while others provide immediate and deterministic finality.

3. Consensus in Peer-to-Peer (P2P) Networks

In a **peer-to-peer network**, all nodes are equal and communicate directly with each other without a central server. Transactions and blocks are broadcast across the network, and each node independently verifies the information.



Hub-and-Spoke: Single point of failure

Mesh Web: No central authority

Consensus is achieved when nodes collectively agree on which transactions are valid and which block should be added to the blockchain. This distributed topology improves fault tolerance, decentralization, and resistance to censorship.

4. Common Attack Models

One common attack is the **Sybil attack**, where an attacker creates many fake identities to gain influence over the network.

A **51% attack** occurs when an attacker controls the majority of the network's computing or staking power, allowing them to manipulate transactions.

A **double-spending attack** involves spending the same digital asset more than once, which consensus mechanisms are designed to prevent.

An **eclipse attack** isolates a node by surrounding it with malicious peers, causing it to receive false or limited information.

A **denial-of-service (DoS) attack** aims to disrupt the network by overwhelming nodes with excessive traffic.